MAST32001 Computational Statistics

Exercise Sheet 6 – Gibbs sampling, HMC, Stan, and variational inference

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WS 2026

Instructions

Submission

Solutions must be submitted in **Moodle no later than 14.10.2026 at 12:00**.

Please submit your solutions in the **submission area corresponding to your exercise group**. Make sure that you submit to the correct exercise group.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point(s) for a task, you must:

1. submit your solution by the deadline;
2. attend the exercise session in which the task is discussed; and
3. be willing to **present, explain, and discuss your solution** during the exercise session if asked.

Submitting a solution without attending the corresponding exercise session does **not** earn exercise points.

Your solution does not need to be perfect in order to be discussed. The purpose of the exercise sessions is also to compare approaches, identify difficulties, and learn from different solutions.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted**.

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work**.

Exercise 1: Gibbs sampling with a latent variable

Suppose

$$Y_i = \theta + Z_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where

$$Z_i \overset{iid}{\sim} N(0, \tau^2), \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2),$$

and

$$\theta \sim N(0, s_0^2).$$

Assume that τ^2, σ^2, s_0^2 are known.

1. Derive the full conditional distribution of each Z_i given θ and y_i.
2. Derive the full conditional distribution of θ given $Z_1, \dots, Z_n$ and the data.
3. Implement a Gibbs sampler in R.
4. Simulate a data set with $n = 50$, $\theta = 2$, $\tau = 1$, $\sigma = 0.5$, and $s_0 = 5$. Run the sampler and estimate the posterior mean and standard deviation of θ.
5. The latent variables are not actually needed for this model. Integrate out Z_i and identify the marginal sampling distribution of $Y_i \mid \theta$. Explain one advantage and one disadvantage of introducing the Z_i's anyway.

Exercise 2: HMC and the geometry of a correlated target

Consider the bivariate normal target

$$\theta = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma \right), \qquad \Sigma = \begin{pmatrix} 1 & 0.98 \\ 0.98 & 1 \end{pmatrix}.$$

Ignoring additive constants,

$$\log \pi(\theta) = -\frac{1}{2} \theta^\top \Sigma^{-1} \theta.$$

1. Derive $\nabla_\theta \log \pi(\theta)$.
2. Plot contours of the target. Explain why independent random-walk proposals in the two coordinate directions are poorly matched to its geometry.
3. Write a function in R that returns the log target and another that returns its gradient.
4. Transform to
$$u = \frac{\theta_1 + \theta_2}{\sqrt{2}}, \qquad v = \frac{\theta_1 - \theta_2}{\sqrt{2}}.$$
Calculate $\mathrm{Var}(u)$, $\mathrm{Var}(v)$, and $\mathrm{Cov}(u, v)$.
5. Use your result to explain why reparameterization or a suitable HMC mass matrix can substantially improve computation.

Exercise 3: Variational approximation and a missing dependence

Suppose the target distribution is

$$p(\theta) = N_2(0, \Sigma), \qquad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix},$$

with $|\rho| < 1$.

We approximate it by a mean-field family

$$q(\theta_1, \theta_2) = q_1(\theta_1) q_2(\theta_2).$$

1. Starting from the CAVI identity

$$\log q_j^*(\theta_j) = E_{q_{-j}}\{\log p(\theta)\} + \text{constant},$$

   derive the optimal forms of q_1 and q_2.
2. Show that at the CAVI fixed point both variational means are zero.
3. Find the variational marginal variances.
4. Compare these variances with the true marginal variances. What happens as $|\rho|$ approaches one?
5. Verify the result numerically in R for $\rho = 0,\ 0.5,\ 0.95$.

Exercise 4: MCMC or variational inference?

A researcher fits a Bayesian logistic regression with $p = 40$ predictors to a large data set. Two computational analyses are available.

Analysis A: four NUTS chains, each with 1000 post-warmup draws. The chains have $\widehat{R}$ close to 1, satisfactory effective sample sizes, and no divergent transitions. Runtime is 25 minutes.

Analysis B: a mean-field variational approximation. The ELBO stabilizes quickly and repeated fits from different initial values give nearly identical ELBO values. Runtime is 20 seconds.

For one coefficient, the summaries are

	mean	SD	$P(\beta > 0)$
NUTS	0.31	0.22	0.92
VI	0.30	0.13	0.99

For a second coefficient, both methods give almost identical means and standard deviations.

1. Explain why a stable ELBO does **not** establish that the VI approximation is close to the posterior.
2. Give a plausible explanation for the smaller VI standard deviation in the table.
3. Both coefficients come from the same variational fit, yet the approximation is accurate for one and not the other. Give a plausible explanation.
4. If the scientific goal is accurate posterior uncertainty for a small number of coefficients, which additional checks would you perform before using the VI results?
5. If the goal is to fit thousands of similar models for exploratory screening, explain why VI might nevertheless be useful.
6. Write a short paragraph explaining why the question "Is MCMC or VI better?" has no universal answer.

Exercise 5: Norppa course feedback

Please complete the **Norppa course feedback form** for MAST32001 Computational Statistics:

Open the Norppa feedback form

The feedback form will be available starting **13 October 2026**.

For this exercise submission, include a screenshot showing the Norppa feedback form **before you fill it out**. The screenshot is sufficient as proof of participation; you do **not** need to reveal your answers or submit a screenshot of the completed form.

Requesting a screenshot from before completion ensures that your actual feedback remains private.